Program code:

(a) The transfer function from the given state model

```
clear;

clc;

xdel(winsid());

A=[0 1 0;0 0 1;-5.008 -25.1026 -5.03247];

B=[0;25.04;-121.005];

C=[1 0 0];

D=[0];

disp('The given state model: State equation x*=Ax+Bu and Output Equation Cx+D
where');

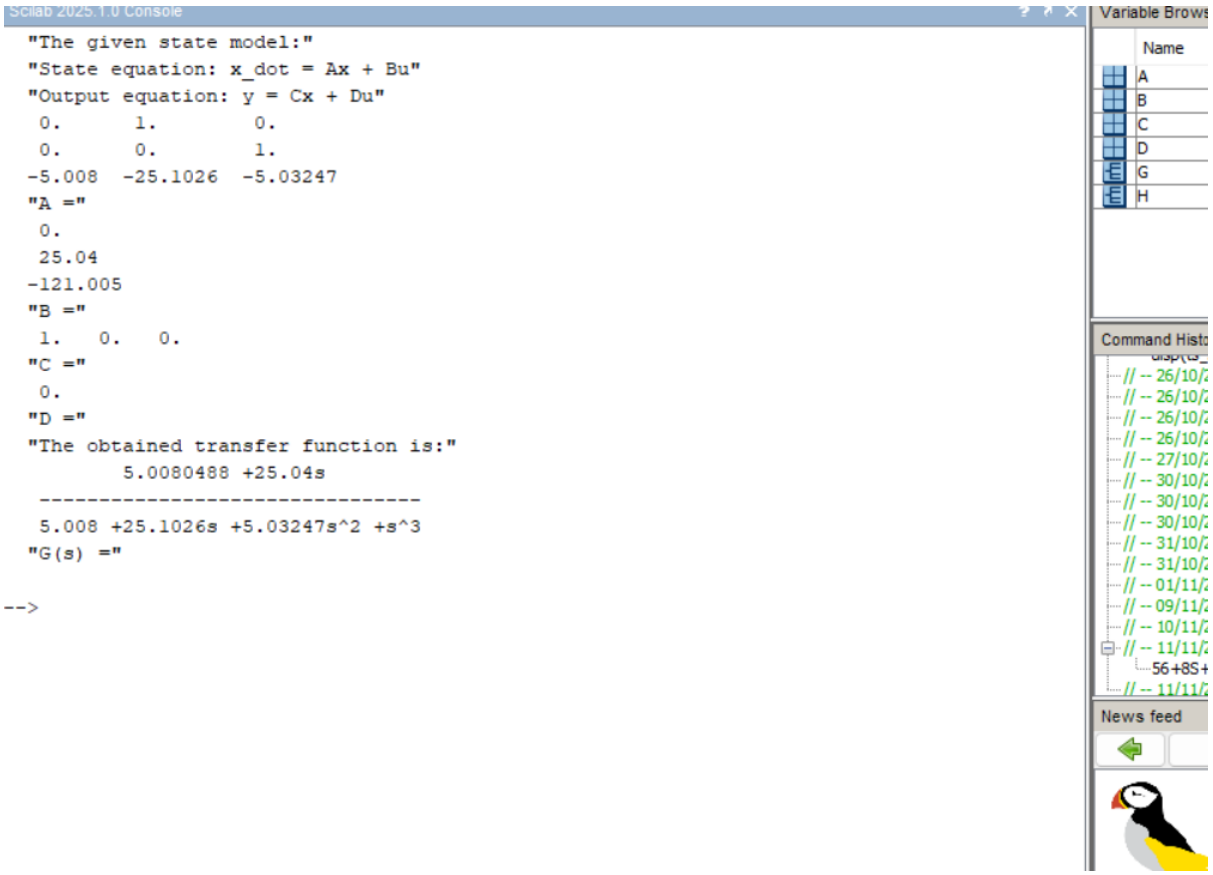
disp(D,'D=',C,'C=',B,'B=',A,'A=');

H=syslin('c',A,B,C,D);

G=clean(ss2tf(H));

disp(G,'The obtained transfer function is G(s) = ');
```

OUTPUT:



(b) The state model from the given transfer function

Program code:

```
clear;

clc;

xdel(winsid());

s=%s;

Htf=syslin('c',s,160+56*s+14*s^2+s^3);

Hss=tf2ss(Htf);

disp(Hss,'The obtained state space model for the given transfer function: State Equation

x*=Ax+Bu, Output Equation Y=Cx+D');

//To print the answer, Use

//ssprint(Hss)

//Alternate:

[A,B,C,D]=abcd(Htf)
```

//To cross check, Use

//Htf2=clean(ss2tf(Hss)) //which matches with Htf

disp(D,'D=',C,'C=',B,'B=',A,'A=');

OUTPUT:

